

Align

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1 Background

Anyone who has participated in the fitness community for any substantial period of time can tell you that the fitness community has changed dramatically over the span of the last few decades, let alone few years. This has included increasing scientific rigor that has been and continues to be invested to discovering optimal methods of reaching different fitness goals. Long ago are the days of anecdotal training methods; "muscle magazines" promising to help you arrive at your fitness goals before the start of the summer. This scientific rigor has not only been exclusive to laboratories and kinesiologist and sports trainers, but has begun to trickle-down to general populace. This phenomena has become so prominent it has even begun to be known somewhat pejoratively as "evidence/science based lifting".

One subsection of this enterprise is in **weighted movement patterns**. The question being asked is: what is the most optimal pattern of movement of a human body under a load that maximizes performance metrics while

minimizing injury(i.e. longevity of performance)? This question is especially pertinent to the increasing body of gym-goers whose goals are to build musculature, increase strength and power outputs, all while being injury free. Though discovering and teaching these optimal movement patters has long been a matter of much difficulty. We now take for granted that prior to the internet the scope of health related information was limited, laden with misinformation and folk lore that could neither be denied nor confirmed. With the growing body of information, we are now discovering what is fact and what is fiction, and reforming year by year thoughts in general fitness that were thought to be common sense.

What investigation has discovered is that there in fact does exist optimal patterns of movement under certain loads. That is, there are better and worse ways of performing basic patterns of movement with weights applying resistance to the body. Not only that, but that there are certain basic movement patterns which when done under load are more conducive to health and fitness metrics than others. In the positive direction is achievement of fitness goals, in the negative direction is at best failing to meet said fitness goals, but at worst getting injured during performance undermining the entire enterprise.

As shown by the health and fitness association, participation in health club is at a all time high(in particular in the tech savvy Gen-z populace), and of which orient their exercise routines primarily around free-weight movements[1].

2 Description

Align is a mobile based application that attempts to guide user to optimal performance of weighted movement patterns through the usage of artificial intelligence technology. The weighted movement patterns of interest are those considered most rudimentary, beneficial, accessible, and widespread among not only the general populace but sports trainers as well. A general survey finds that would tend to be the squat, bench, and deadlift. While the initial version of the app aims for being able to train a user in these exercises, it seems rather trivial to extend the principles used to implement these exercises to any other exercise of choice. The principles and heuristics that allow implementation are as followed.

We assume that there exist some optimal performance of a movement pattern under load and that it can be defined in bio-mechanical terms. This assumed optimal performance will not only allow quantitative comparison but serve as the point of reference for which the user will be measured against. There will naturally be variability in each individual's ability to perform an exercise optimally. Factors like bodily leverages, flexibility, etc. are limiting factors in a user's ability to perform an exercise as defined optimally, sometimes in even irreconcilable ways. This will be optimally be taken into account when assessing performance of an exercise and giving feedback¹.

Not only would a comparison be made, but also and most importantly, the points of failure, or most probable cause for this lack of alignment with the optimal performance model should be assessed and reported. This is

¹See ethical consideration for more.

the critical feature that would allow users to show improvement over time; that and a system that would report fault, but not show possible areas of improvement might be liable to misdirect the user as to the origin of these faults.² The quantitative comparisons alongside critiques of the performance, if availed by the user, as they would a personal trainer-which is in fact what is being simulated-should result in improvement in performance over time. This improvement over time should be available as visible feedback. Direct replays and comparisons, of different performances for review would also be optimal in terms of improving over time. Again, this is all under the assumption that the faults that would cause misalignment exist and can be described in bio-mechanical terms.

3 Implementation

3.1 Machine Learning & Computer Vision

For our machine learning model, we used a basic supervised machine learning algorithm. This model would take as input, the measurement of angles between points on the body, specifically the joints. We broke down the performance of a squat as a series of points within the interval of standing, hitting full squat depth, and standing back up again. These measurement of angles at certain points will be known as snapshots at a particular moment in the movement. The machine learning model does not judge the measurement of these angles as a whole, as in the entire set of angles at one time, but rather judges each particular instance of a snapshot. It thus has a set of

²Again this will be taken up in the ethical consideration section

weight, stored in a JSON file, that apply to said series of snapshots. Each particular snapshot is either seen as a success or a failure. We determine the accuracy of the whole performance, by taking the percentage of the number of screenshots determine to be successful.

In order to capture these exercise performances we implemented a real-time computer vision solution using the Media Pipe API. It contains a computer vision application, that allows for capturing, processing, analyzing, and displaying body position and movement in real time. It follows that processing and subsequent calculation of angles should only be done during performance of the movement. In turn we defined a certain measurement of angles between joints as the start and end-point of the movement(given that the start and end-point are identical). To begin performance recording a user would have to break out of the start-point, and enter back into it, this would be considered one repetition; at which point the user may choose to end the recording or input another repetition.

3.2 Database System

There necessarily must exist a database which holds user information and a history of performances associated with a user, alongside the metrics which determined those performances. For the time being, only the relation containing user information would have to be hosted, while performances can be kept locally. Although we used FASTAPI to implement and interact with one remote database which contained both of these relations. This database will be necessary, for displaying and giving user feedback on performances as well as sign-in and access control capabilities. The training data-and thus

also the user metrics for performance-would be stored as a tuple of angles, each being a angle created between a pair of points on the body. A repetition, a set of tuples of angles over a interval of time, would be classified into either a good repetition, or a bad repetition and that is how our training data will be formed.

3.3 Platform Deployment And The Front End

We decided to deploy the application on android for the time being considering the necessity to record videos in a commercial gym setting. We choose android over ios because of its majority market share holding. Given the platform being deployed on we choose to implement our front end via Kotlin, specifically using the Jetpack Compose UI framework. This front-end includes basic user necessities such as a landing screen, a log-in page, exercise selection, etc. In addition this front end includes graphical displays that serve as feedback for user performance; in particular, a graph of performance over time. As well as and primarily, the video capturing component which displays a pose overlay over the user as they perform the exercise.

4 User Manual

4.1 Log In

Upon opening the application, you will be prompted for your username and password. Enter your corresponding credentials and click the log-in button. If you do not have a corresponding account please create one

4.2 Account Creation

If you do not have a account, or wish to make a new account, there exist a sign-up button on the log-in page. You will be prompted for a username and a password. The password cannot already exist in our database, that is the password must be unique. Passwords and usernames are limited to alphanumerical characters. Upon filling out these forms, pressing the sign-up button, will attempt to create a account with the credentials you filled. If the password is unique, and passed validation, your account will be created and you will redirected to the main page

4.3 Altering Account

Displays two prompt for altering username or password. What credentials will be attempted to be altered will depend on what prompt field is left empty. Upon entering info onto prompt field, you can click a change-account button, which will attempt to change credentials. It is done successfully, you will receive a notification that the corresponding credentials have been changed. Otherwise, you will receive a prompt that the username collides with another user, or that the newly inputted password is equivalent to the one before.

4.4 Exercise Selection

Upon being successfully logged-in, you will enter into the exercise selection page, which also serves as the main page. From this hub, you could either choose one of the displayed exercises to get your performance analyzed, there

will exist widgets on hub page that will take you to either the account settings page, where you change your account credentials, or the performance history page, where you can get graphical feedback based on your history of analyzed performances. If an exercise is selected, then you will be directed to the performance recording page where you may record your performance of an exercise and receive a subsequent rating.

4.5 Performance Recording

Recording of performance will begin automatically, simply get into position, and perform the associated exercise for as many reps as willing. upon completion, simply press the end-recording button displayed on the screen, to end exercise performance. After which, a rating for your performance will pop-up, analyzing overall performance as well as each individual rep. This performance rating can be re-displayed for analysis in the performance history page

4.6 Performance Graph

There exist a widget on the main page that once pressed on, allows you to access a history of performance judgments. You are prompted to enter a interval for which you would want displayed. You will be shown a shaded graph with points that constitute this graph. You can click on any one of these points to see further specifics on the metrics of the performance.

5 Roadblocks

There was difficulty in properly using tools that facilitated deployment of software packages across different systems and management of said software package. That is, we struggled with the **devops** of developing software. We came across various difficulties when using git and gitlab, with various instances pipeline failures, git merge request failing due to conflicting commits and various sorts of other problems that come from version control tools. In addition, we used the Gradle build tool for developing in Kotlin, a tool and language which all of the team were unfamiliar with. This resulted in various builds scripts being committed that would break compilation on one system or another. These rather inconsequential cases would end up costing the team a significant amount of time.

Conceptualizing the usage and methods of tracking and judging a repetition of an exercise proved to be not so difficult, but it was rather the implementation of these concepts that proved time consuming. As in, it was very easy as a human to have these intuitive understanding of these concepts, but translating that into mathematical and abstract terms such that a computer could work about these concepts proved interesting. Some such instances were questions such as: how do we identify the start and end of a repetition? How do we translate the performance of an exercise into data for a model to work on? How do we classify a good squat v.s. a bad squat?

Lastly is the general skillsets and understandings of development tools and apis that allow us to implement the software package. These include things like authentication and retrieval of information scoped by user cre-

dential

6 Ethical Considerations

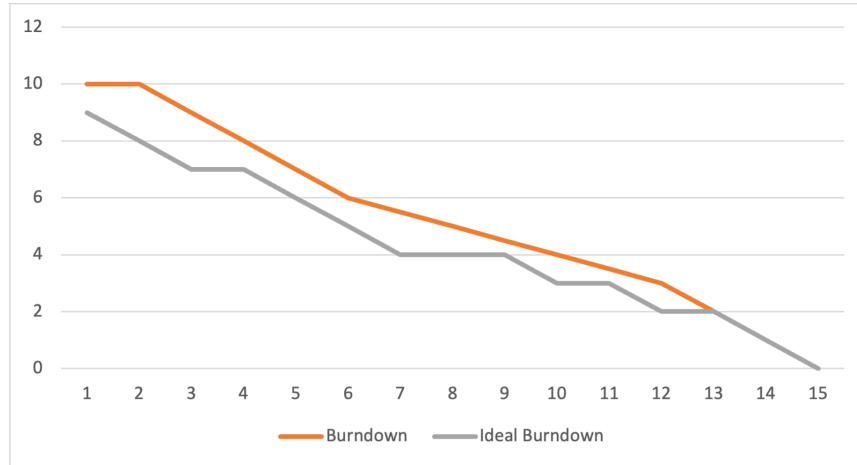
The most blatant possible ethical violation stems from the fact that incorrect performance of exercise, especially under heavy loads, can and does often lead to injury. If the advice given by the model is incorrect, the application could very well be held liable for the injury. There may be various sources from which poor or faulty advice of exercise performance may be given by the model. One factor is the fact that human morphology varies widely from individual to individual, and hence also their physiology differs. This is more of a consideration for exercise science in general of whether there exist a universally correct performance of an exercise given this variance in phenotype. The current consensus on optimal performance may or may not be skewed for a certain phenotype rather than others. While some of these anatomical features are observable and can possibly be taken into account when weighing the performance of an exercise, many are not. Anatomical variation can occur both interpersonally (between individuals) and intrapersonally (within a single individual), e.g. one hip joint may not connect to the hip socket in the same way that the other hip joint does in the very same person. The issue comes up in not only judging performance but also giving advice for improvements. If the origin for some failure is anatomical nature and there exist no possible neuromuscular adaptation that can take place to improve scores, this could very well lead to injury as well.³ For the

³Take a look at [2] for more on anatomies effect on exercise performance

sake of the application we will take it as satisfactory the assumption that given a large enough group sample, variance in morphological features will form a bell curve. We will also have to warn the user against the biases with which the model may be in-built with and the limitations of current scientific literature.

The other ethical considerations are those that are general to any software application and unrelated to general exercise science. Which includes but is not limited to integrity, confidentiality of user information and the like. It is not out of the picture that a stock of videos of subjects performing a set of exercises, especially a sorted and tagged stock of such videos may be of value. Such data could be used in the training of other models or for research purposes. In this case, we at Align would not necessarily be opposed to allowing access to the store of these videos under the stipulation that users have acknowledged some terms and conditions that notify them of this possibility and their agreement to do so.

7 Scrum Process



As is betrayed from the burndown chart the pattern of development was one in which we were consistently lagging behind expectations. Especially as we begun development, we found ourselves having to carry over user stories into successive sprints. This is especially where we struggled with devops and working together in general as a team. As we approached deadline though we increasingly approached matching our ideal burndown. Devops issues were gotten under control and soon the only problems left were ones of development and implementation. At times it was rather questionable whether we would "make it in time" but nonetheless we ended up aligning with our goals.

8 Contribution

Our group members use their various expertise to contribute to various implementations as suited to their abilities and capabilities. Working primarily

on the front end was Jordan, who helped in configuring and implementing the UI of the mobile application. This included things like the log-in pages, the exercise selection pages, as well the graphical display of past performances. As well beyond the front end also assisted in establishing the schema for the database and how data would be classified. Justin was the primary back-end developer and was responsible for things such as implementing the computer vision capabilities through the media pipe API, establishing connections between the frontend and backend, the functionality to create the history chart, connecting the frontend to the database, and integrating camera features. Radion was primary responsible for developing the machine learning model that was used in judging exercise performances; alongside some additional front end work in the form associated with user logins, as well as some work on the password systems and the video hosting capabilities. Finally, Wilbert provided much of the concepts and organization of the application that inspired implementation through knowledge of software engineering and exercise science. In addition worked on features of the application such as management of account credentials, devops operations in the form of resolving issues that came up during merge request and contributed to the machine learning model in the form of some fast-forwarding functionality on the mobile application level

9 Lessons Learned

The general process-as is the goal of the class-has been one about learning principles of software engineering, and especially why these methods were

developed in the first place and why they work. Though it is simpler to work on a software system on ones own, the general potential of this project given only the workload of one developer is rather limited. It is only when more than one developer begins to work on some software package that that software can begin to be of note and hence also be marketable. The difficulty with these larger groups of developers is that the development process and the transfer of information undergoes entropy, i.e. becomes more chaotic over time. This is exactly when methods of software engineering becomes critical for ensuring the integrity and development of that software. It becomes easier to appreciate and see the purpose of these philosophies and tool sets of software engineering, especially mastery of these tool sets.

In addition, we have learned about the complexity and nature of software systems in general. We often take for granted the complexity that underlies software applications. The pre-packaged unity of a application presents itself before us and we take it as it is; not knowing the unfathomable sum of parts and the their interactions which ironically allow us to see this facade of wholeness. It transformed standards and idioms from pedantic encumbrances into universal conveniences. That individual styles must be sacrificed for the sake of convenience and scale. Poignantly put: we have learned more about the relations(links) that makes possible the communication between differing processes.

Finally, we have learned about the processes itself.

10 Future Work

As of currently, we have only managed to implement a model for the performance of a squat, and only a good or a bad squat. Though the underlying training methodology can be used to implement any set of exercises with relative ease. We would also like to train the model to identify and classify more of the different ways in which a squat could be incorrect such that specific advice for improvement could be given and expand that methodology to other exercises. This could include breaking down a exercise into a set of contiguously distinct phases, and analyzing performance at those difference phases. Additionally, we would like to spend more time in improving the user interface experience for the user; as of currently we at Align find it underwhelming to say the least. An additional burgeoning possibility is to use user input data in training the machine learning model. Given the application is capable of taking some proportional information about the user, this user metadata may be used in creating models that are more adjusted for different body types. Additionally the methodology implemented for this application could be expanded, as mentioned before, into arenas of physical movement outside of loaded movement patterns.

References

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